

Three separately frozen experiments | qualitative synthesis only

No pooled population | no experimental data flow between panels

Study 3

Temporal runtime evidence

Gate-visible evidence

Signature validity; freshness;
received authorization evidence;
contact-dependent record
availability; security signal.

Research-only truth

Hidden authorization truth.

Residual boundary

Fresh, validly signed V5 evidence
can remain false; a truthful
pre-onset cache can briefly lag a
state change.

Effect of stronger composition

K4 restriction reduces selected
modeled exposure but does not
eliminate persistent V5
qualification for B0/S1.

Study 4

Producer composition

Gate-visible evidence

Signed producer claims; vote
threshold; synthetic
provenance-domain count.

Research-only truth

Hidden authorization truth.

Residual boundary

Same-size compromised subsets can
differ because their
provenance-domain composition
differs.

Effect of stronger composition

Provenance can delay systematic
unsafe qualification, cause
earlier benign rejection for
selected subsets, and have null
threshold effects.

Study 6

Recovery-artifact assurance

Gate-visible evidence

Signature; digest; provenance;
reproduced-build match;
source-review attestation;
release approval.

Research-only truth

Objective baseline correctness.

Residual boundary

APPROVED_BAD_SOURCE remains
qualified when all six
gate-visible assurance signals
are true.

Effect of stronger composition

Additional signals close
specified modeled states while
increasing sensitivity to benign
assurance-signal loss.

Parallel panels are a qualitative manuscript synthesis, not an integrated recovery architecture. Only Study 3 models contact.